Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Hydrocarbon-based High-temperature Ion-exchange Membranes for Electrochemical Hydrogen Compressor Application			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (**V**), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Chemicals	
	Medium Category	Medium Category	Medium Category	Small Category
	New & Renewable Energy	New & Renewable Energy	Polymeric Materials	Materials for Energy/Environmental Industries
Concept of Proposed Technology	Fabrication of high-temperature electrochemical hydrogen compressor devices using hydrocarbon-based ion-exchange membranes for hydrogen storage and transportation.			
Rationale for Joint R&D Project	The RPI team has developed a novel class of hydrocarbon-based ion-exchange membranes tailored for use in high-temperature electrochemical hydrogen compressors (EHCs) to reduce the overall cost of hydrogen transportation and storage by leveraging the current infrastructure of natural gas pipeline network. To advance this technology, a joint R&D project is proposed to design and test membrane electrode assembly (MEA) devices, with the ultimate goal of achieving commercial viability.			
Key Words	1. hydrogen 2. Hydrogen storage and transportation 3. electrochemical hydrogen compressor 4. ion-exchange membranes			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

Summary

This proposal aims to translate the hydrocarbon-based anion-exchange membranes developed at RPI into commercial high-temperature electrochemical hydrogen compressors (EHCs) for hydrogen storage and transportation, in collaboration with a small or medium-sized enterprise (SME) in Korea.

Technology Demand

The cost reduction of hydrogen is a key challenge for transitioning the energy system from carbon-based to hydrogen-based one. In the past decade, the cost-reduction efforts for hydrogen have primarily focused on the production and use of hydrogen, such as electrolyzers and fuel cells, but more than 60% of the hydrogen cost comes from transportation and storage, because they rely on distribution by trucks and dedicated storage and refueling stations [1,2]. This proposal claims that hydrogen storage and delivery costs can be significantly reduced through pipeline delivery, down to ~10% of truck delivery and handling costs. By reducing delivery and storage costs, hydrogen can achieve a significant cost advantage; its cost can be less than 50% of oil. However, the initial capital investment for the hydrogen pipeline network has been prohibitively expensive.

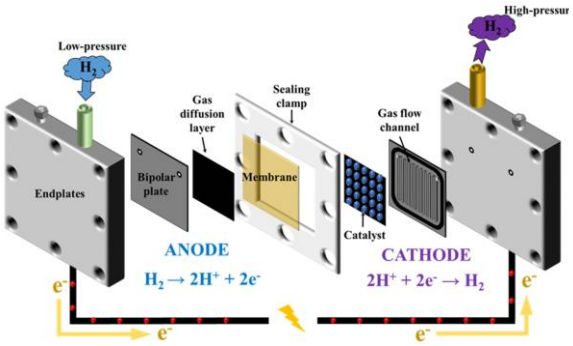
Instead of constructing dedicated hydrogen pipelines and storage facilities, leveraging the existing natural gas pipeline network for hydrogen storage and delivery can significantly reduce capital costs and accelerate the establishment of the hydrogen economy. In this strategy, hydrogen is blended into natural gas up to 20% and transported to the target local storage and refueling station through the natural gas pipeline network. For example, South Korea's natural gas pipeline network, which runs more than 5,000 km, has an estimated hydrogen storage capacity of approximately 500,000 kg, sufficient to charge 100,000 hydrogen fuel cell cars. This strategy can be employed for any country with an existing natural gas pipeline network, and significant market opportunities for associated devices and technologies are expected.

Natural gas pipeline network of South Korea for hydrogen transportation and storage (Image from KoGas Co.)

Proposed Technical Approach

In this strategy, the key technical requirement is the efficient, low-cost separation of hydrogen from the hydrogen-natural gas mixture in the pipeline, followed by hydrogen compression for high-pressure refueling. The proposed development and commercialization of EHCs using RPI's ion-exchange membrane will be a leading technology for hydrogen storage and transportation via the existing natural gas pipeline network, as described by the following technical advantages.

The membrane electrode assemblies (MEAs) of EHCs are very similar to those of fuel cells and electrolyzers, but operate in the reverse direction [3]. However, the ion-exchange membrane requires different properties for efficient and stable operations of EHCs as follows, in particular for the high-temperature phosphoric-acid-based EHCs (HT-PA-EHCs) [4]:



Membrane electrode assembly diagram of EHC [4].

- High thermal resistance for the EHC operation at 140 – 160 °C for the following requirements.
  - Simplified water management because the proton transfer proceeds using phosphoric acid, and the high temperature minimizes water flooding that causes loss of phosphoric acid.
  - Reduced mechanical stress on the membrane due to the lack of water swelling and drying cycles.
  - High tolerance of Pt catalyst poisoning at the operating temperature, in particular by CO that is generated from hydrocarbon-based cracking, such as grey hydrogen by steam methane reforming.
  - High-temperature operation accelerates the mass transport rate of protons and separation.
- Strong mechanical properties for minimal creep behavior due to the large pressure difference between the anode (hydrogen and natural gas mixture) and cathode (compressed pure hydrogen) at the high-temperature condition.
- Strong chemical inertness for the reactive species, such as radicals from the Fenton reaction.

Current State of EHCs and Proposed Solution

EHCs have been fabricated using phosphoric acid-doped polybenzimidazole (PA-doped PBI), but retaining PA in PBI during operation has been challenging due to the weak basicity of PBI to PA [4,5]. Recently, it has been found that anion-exchange membranes with quaternary ammonium functional groups exhibit excellent retention of doped PA due to strong ionic interactions between PA and the basic quaternary ammonium groups. However, for the high-performance HT-PA-EHCs applications, the anion-exchange membranes should meet the physical and chemical requirements described above. RPI recently developed a new class of hydrocarbon-based anion-exchange membranes tailored for high-performance HT-PA-EHCs: quaternary-ammonium-functionalized polymer, which is referred to as QA-BD, shown in Figure 1b.

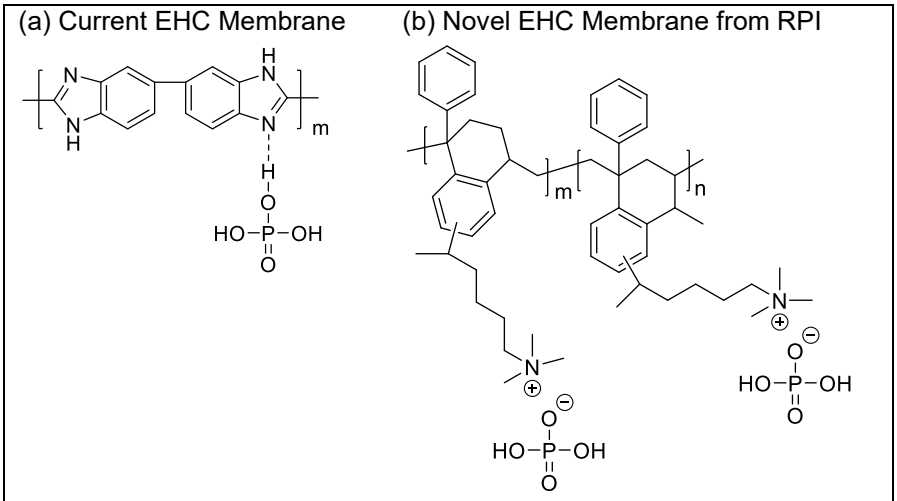


Figure 1. (a) Polybenzimidazole doped with phosphoic acids. Phosphoric acids, serving as proton carriers, are easily leached due to the weak interaction between phosphoric acid and imidazole. (b) Quaternary-ammonium-functionalized polymer (QA-BD) salted with phosphoric acid. The ionically bound phosphoic salts to the QA-BD polymer are effectively retained in the membrane and conduct protons in EHC devices.

Target R&D Objectives

The proposed research aims to enable the development and commercialization of low-cost and high-performance HT-PA-EHCs for hydrogen transportation and storage by leveraging the existing natural gas pipeline network. The proposed grand objective will be achieved through the following specific aims.

At RPI,

- A series of high-temperature ion-exchange membranes will be designed and synthesized for HT-PA-EHCs applications as described in the R&D research plan below.
- The synthesized membranes will be characterized for mechanical strength, thermal stability, and conductivity at the operation conditions of HT-PA-EHCs to meet the target specifications.
- Properties of ion-exchange membranes will be tuned and improved by molecular design and membrane processing through active communication with the partner SME.

At the partner SME,

- Fabrication of the MEAs of HT-PA-EHCs based on the RPI's ion-exchange membranes.
- Test the MEAs under in-lab and the operating conditions at the natural gas pipeline.
- Provide feedback to RPI for improvement of the ion-exchange membranes.

Those target R&D objectives aim to develop and commercialize competitive, high-performance EHC devices for the hydrogen transportation and storage market.

R&D Target Specifications

The proposed research will achieve the following objectives and specifications for ion-exchange membranes and HT-PA-EHCs to be developed in collaboration with the partner SME.

Property	Temperature	Relative Humidity	Target Specification
Conductivity, Phosphoric-acid-doped membrane	50 °C	60 %	Greater than 25 mS/cm
	160 °C	0 %	Greater than 100 mS/cm
Conductivity Stability	10 Cycles from 80 °C to 160 °C	N/A	Maintains greater than 80% of the maximum
Weight Loss, Phosphoric-acid-doped Membrane	60 °C for 50 hours	40 %	Less than 10 %
	160 °C for 50 hours	0 %	Less than 25 %
Strain at Break	80 °C	40 %	Greater than 1 %
	160 °C	0 %	Greater than 2 %
Young's Modulus	80 °C	40 %	Greater than 100 MPa
	160 °C	0 %	Greater than 30 MPa

R&D Plan

Ion-Exchange Membrane Materials for High-Temperature Applications: Synthesis and Properties

RPI will synthesize and conduct primary tests of the QA-BD ion-exchange membranes for HT-PA-EACs, molecularly designed for high-temperature and chemically harsh operating conditions.

The preliminary conductivity test of hydrated QA-BD membrane with Cl^- counter-ions demonstrates excellent performance and tunability for HT-PA-EHC applications (Figure 2). Two QA-BD ion-exchange membranes of different ion-exchange capacities record a conductivity linearly increasing with temperature for the membrane of 1.9 meq/g (lower blue curve in Figure 2) and a conductivity drop at 80 °C for the 2.4 meq/g due to the high ion-exchange capacity causing macrophase separation of the membrane (upper orange curve in Figure 2). This result suggests that the ion-exchange capacity (IEC) of QA-BD can be adjusted over a wide range to meet target applications and performance requirements. In the HT-PA-EHC applications, the QA-BD membranes are solvated with PA, which has a different solvation affinity than water; thus, the demonstrated IEC tunability enables tailored molecular design of QA-BD for target operating conditions with PA.

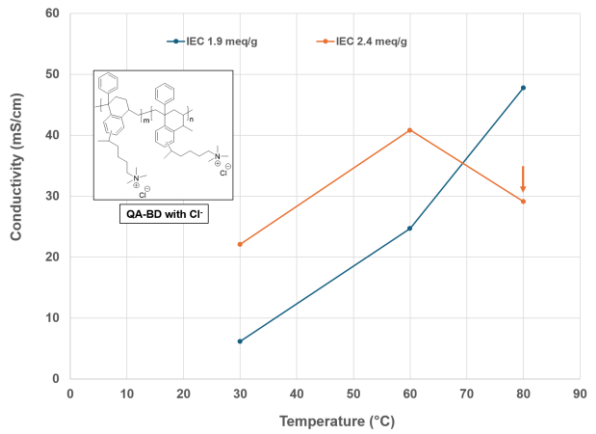


Figure 2. Conductivity of fully hydrated QA-BD with Cl^- counter ions at 30, 50, and 80 °C. The conductivity drop of QA-BD due to excssive water swelling is marked with a down-arrow.

These preliminary results demonstrate the applicability of the QA-BD membrane for HT-PA-EHC devices, which will provide an innovative solution for the hydrogen storage and transportation problem by leveraging existing natural gas infrastructure. Detailed R&D plans are followed.

Phase 1 Timeline and Milestones (0 - 0.5 Yr)

RPI will demonstrate the synthesis and preparation of QA-BD membranes saturated with phosphoric acid for HT-PA-EHCs applications (Figure 1b). The QA-BD membrane will be sent to the partner SME for the fabrication of a prototype HT-PA-EHC device and the following tests.

Timeline	Milestones
0 – 0.25 Yr	● RPI completes the synthesis and preparation of a set of initial QA-BD membranes and property measurements. ● RPI sends the QA-BD membranes to the partner SME for the fabrication of a prototype HT-PA-EHC device and initial test.
0.25 – 0.5 Yr	● RPI tunes the QA-BD polymer and improves the membrane preparation process based on the discussion with the SME partner. ● RPI plans and prepares the QA-BD membranes for Phase 2.

Target Specifications in Phase 1

Property	Temperature	Relative Humidity	Target Specification
Proton conductivity, Phosphoric-acid-doped membrane	50 °C	60 %	Greater than 20 mS/cm
	80 °C	40 %	Greater than 40 mS/cm

Phase 2 (0.5 – 3.0 Yr)

RPI will advance the QA-BD membranes through molecular engineering to meet the "R&D Target Specifications." Perhydropolyarene motifs will be incorporated into the QA-BD polymer to improve mechanical, thermal, and chemical stability (Figure 3). RPI will prepare a series of reference QA-BD membranes and down-select membranes for the HT-PA-EHC device fabrications with the partner SME. The partner SME will conduct in-lab and semi-field tests on fabricated EHC devices, and RPI will fine-tune the membrane to meet or exceed the target specifications. RPI will work with a subcontractor of the partner SME to scale up material synthesis and membrane preparation for the commercialization of the HT-PA-EHC devices developed.

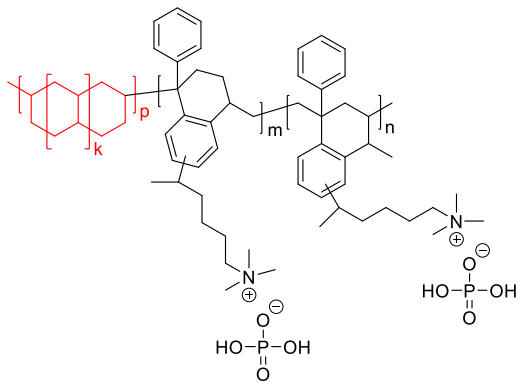


Figure 3. Quaternary ammonium functionalized perhydropolyarene BD polymer to be employed in Phase 2.

	Timeline	Milestones
	0.5 – 1.0 Yr	● RPI prepares a series of QA-BD membranes and down-selects based on conductivity and associated physical properties. ● RPI sends the selected QA-BD membranes to the partner SME for the fabrication of the HT-PA-EHC devices and in-lab test. ● RPI documents membrane casting conditions, *e.g.*, solvent compositions, temperature, drying rate, for membrane performance enhancements. ● RPI conducts characterization of QA-BD membranes to improve membrane performance. ● RPI actively communicates with the partner SME to solve problems and improve the performance of membranes and HT-PA-EHCs.
	1.0 – 2.0 Yr	● RPI works with the partner SME to prepare and conduct the hydrogen separation from high-pressure natural gas and hydrogen mixtures, a semi-field test. ● RPI designs and fabricates QA-BD membranes based on the hydrogen separation performance from high-pressure natural gas and hydrogen mixtures.
	2.0 – 2.5 Yr	● RPI will conduct final tuning of the QA-BD membranes to meet or exceed the target specifications for the commercialization effort of the HT-PA-EHC devices using the RPI membrane. ● RPI will assist with the partner SME's scale-up synthesis and membrane fabrication efforts.

4. Role & Responsibility (R&R)

Expected R&R of Research Institute	● RPI will provide high-performance ion-exchange membrane materials to the partner SME for the fabrication of HT-PA-EHC devices and tests. ● RPI will tailor the ion-exchange membranes to maximize the performance and stability for HT-PA-EHC commercialization based on the communication with the partner SME. ● RPI will provide scientific and technical advice to the partner SME for the design, fabrication, and testing of the HT-PA-EHC devices. ● RPI will provide scientific and technical advice for the scale-up synthesis of the membrane synthesis and the membrane casting process. ● RPI will lead and organize regular meeting sessions with the partner SME to achieve the target R&D aims.
Expected R&R of Korean SME Partner	● The partner SME leads the design, fabrication, and testing of the HT-PA-EHC devices based on the QA-BD membranes RPI provides, which will have a range of different electrochemical and chemical properties for the optimization of the HT-PA-EHC devices. ● The partner SME conducts a series of HT-PA-EHC device tests and shares the test results with the RPI for membrane performance improvement. ● The partner SME coordinates the scale-up synthesis and fabrication of the membrane for the readiness of the commercial production of the HT-PA-EHC devices. ● The partner SME conducts the market and policy analysis for the commercial deployment of the HT-PA-EHC devices.

5. Expected Outcome/Impact

Expected Outcomes and Impact	● The HT-PA-EHC devices to be developed using RPI's innovative QA-BD membranes will enable rapid deployment of the hydrogen economy with minimal investment in hydrogen storage and transport infrastructure through leveraging the existing natural gas pipeline network. ● The technology has great potentials of being dominating technology and commercialization as it is deployable to any market with existing natural gas pipeline infrastructure.

6. References

[1] *Annual Technology Baseline - Hydrogen*, https://atb.nrel.gov/transportation/2020/hydrogen.
[2] *DOE H2A Delivery Analysis | Hydrogen Program*, https://www.hydrogen.energy.gov/program-areas/systems-analysis/h2a-analysis/h2a-delivery.
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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization